

SMART CONTROL OF WIND-FED BLDC MOTOR DRIVES THROUGH IOT AND SIX-STEP INVERTER TECHNIQUE

K Subha Sharmini

Department of Electrical and Electronics Engineering,
SRM Institute of Science and Technology – Kattankulathur, Chennai, Tamil Nadu, India.
subha.a.star@gmail.com

R. Vinodha

Department of Electronics and Communication Engineering,
ULTRA College of Engineering and Technology, Madurai, Tamil Nadu, India.
vinodhaphd.2019@gmail.com

T. Porselvi

Department of Electrical and Electronics Engineering,
Sri Sai Ram Engineering College, Chennai, Tamil Nadu, India.
porselvi.eee@sairam.edu.in

K Padmanaban

Department of Computer Science and Engineering,
Koneru Lakshmaiah Education Foundation, Vaddeswaram, Andhra Pradesh, India.
padmanaban.k@yahoo.com

Minakshi Sanadhya

Department of Electronics and Communication Engineering, Faculty of Engineering and
Technology, SRM Institute of Science and Technology, Delhi-NCR Campus,
Ghaziabad, Uttar Pradesh, India.
minakshisanadhya@gmail.com

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Abstract: The Brushless Direct Current (BLDC) motor is extensively used in industrial automation, electric vehicle (EV) propulsion, and renewable energy applications due to its superior efficiency, high speed, rapid response, and operational reliability. However, one notable limitation of BLDC motors is the high transient current during startup, which can lead to rapid battery depletion or reduced lifespan of energy storage systems. To address this challenge, this study proposes an Internet of Things (IoT) enabled six-step inverter-based BLDC motor drive system powered by wind energy. The system incorporates a Proportional-Integral (PI) speed controller implemented through a three-phase six-step inverter to regulate the motor's performance. A wireless sensor network collects real-time operational data, which is transmitted to a cloud-based IoT platform for analysis, visualization, and control. The platform features a graphical user interface that facilitates automated monitoring and control of the motor drive. Simulation results confirm that the proposed IoT-based system, combined with the PI controller, effectively tracks and regulates the speed of the BLDC motor under varying load and wind conditions. This integrated approach demonstrates a promising solution for enhancing the efficiency and manageability of wind-powered energy systems using modern motor control and IoT technologies.

Keywords: Motor control, six-step inverter, internet of things, brushless direct current motor, wind energy.

I. INTRODUCTION

The growing concerns over the environmental impact of conventional energy production have accelerated the development and adoption of Renewable Energy Sources (RES) such as wind, solar, and fuel cells. This shift is further driven by the escalating global demand for electricity, the depletion of fossil fuel reserves, and the rising costs of traditional energy sources like coal, natural gas, and oil. Among various RES, wind energy has gained significant traction and is increasingly being integrated into modern power grids. A critical goal in Wind Energy Systems (WES) is to maximize power extraction across a wide range of wind speeds. This can be effectively achieved using Maximum Power Point Tracking (MPPT) techniques, which ensure optimal turbine efficiency under varying wind conditions.

In a typical WES, power is generated by coupling a wind turbine with an electrical generator, either directly or via a gearbox. The generator's rotor is mechanically linked to the turbine (the prime mover), while its stator delivers power to isolated loads or the utility grid through appropriate power electronic converters. As technology advances, the demand for motors that offer high torque at variable speeds, energy efficiency, adaptability, and minimal maintenance has become increasingly important. Traditional motors, such as Direct Current (DC) motors and Induction Motors (IMs), face limitations in meeting these modern requirements. Although DC motors provide high efficiency due to their direct voltage application, they suffer from high maintenance costs and reduced reliability owing to brush wear and spark generation during commutation. Induction motors, while more durable and requiring less maintenance, typically operate with lower efficiency and generate excessive heat.

BLDC motors have emerged as a viable alternative, combining the efficiency of DC motors with the low-maintenance benefits of IMs. Powered by DC voltage and characterized by a trapezoidal back EMF, BLDC motors rely on electronically controlled commutation. A common and effective technique for BLDC motor control is the six-step commutation method, valued for its simplicity and ease of implementation. For consistent torque and smooth operation, precise timing of commutation is essential, whether using the six-step method or sinusoidal Pulse-Width Modulation (PWM).

This study presents an IoT-enabled monitoring system integrated with a six-step inverter-controlled BLDC motor drive, tailored for wind energy systems. The proposed system offers real-time performance tracking, enhanced control, and a cost-effective solution for WES applications. The article details the system's architecture, software components, and experimental results that validate its capability to efficiently monitor and regulate the BLDC motor's operation. Ultimately, the proposed approach aims to improve the efficiency, reliability, and sustainability of wind energy systems, contributing to a more resilient and environmentally responsible energy future.

II. LITERATURE SURVEY

This section presents a comprehensive overview of relevant research on IoT-based monitoring systems and advanced motor control techniques for wind power generation. A hybrid energy storage system based open-winding motor drive equipped with an efficient energy management system for Electric Vehicles (EV) is presented in [1]. The system optimally utilizes energy from ultracapacitors and batteries, maximizes regenerative braking energy usage, and maintains battery safety, offering superior energy efficiency and control when compared to conventional systems.

In [2], a vector-controlled induction motor drive is used to simulate the dynamic behavior of a wind turbine system. The control algorithm matches the wind turbine's characteristics to study the effects of wind speed variations and validate control strategies. A cost-effective motor control method using a four-switch BLDC configuration with a single current sensor and a modified space vector PWM technique is proposed in [3]. This approach simplifies hardware and reduces costs while improving torque and speed regulation. A comparative analysis of PI and field-oriented control (FOC) methods for BLDC motor drives is presented in [4]. The study found that FOC offers superior torque and speed performance, especially at higher speeds, whereas PI performs better at low speeds. To further reduce torque ripple during sector-to-sector commutation, [5] proposes a modified direct torque control strategy with an additional voltage vector incorporated into the switching table.

The electrical efficiency of traditional six-step and inverted PWM methods for BLDC motor drives is compared in [6]. Results reveal that inverted PWM provides improved efficiency by reducing energy losses. In [7], a novel six-phase inverter producing 60-step output voltages is introduced. The system, based on multiple capacitor-switch bridges, delivers enhanced power quality and reduced harmonics compared to standard three-phase inverters. For real-time condition monitoring in WES [8] describes an Industrial IoT based approach employing various sensors across components like generators and power electronics. Machine learning algorithms enables predictive maintenance by identifying faults and estimating component lifespan. Similarly, [9] introduces a neuro-fuzzy-based IoT energy tracking system for smart grids, demonstrating high fault detection accuracy by combining neural networks and fuzzy logic.

A BLDC motor control and speed tracking system using IoT technologies, specifically the ESP8266 Wi-Fi module and Raspberry Pi, is detailed in [10]. The system allows online monitoring and solar-powered motor operation, contributing to energy efficiency. In [11], a modified PI controller with feedback loops is proposed to enhance speed tracking and dynamic performance in EV motors, outperforming conventional PI controllers. A soft start technique for BLDC motors using a six-step inverter and PI controller is presented in [12], showing smooth start-up and stable operation. An economical and practical solution for wireless three-phase motor monitoring is demonstrated in [13], using Raspberry Pi and Wi-Fi modules along with machine learning for fault detection and predictive maintenance.

In [14], an IoT-based system for industrial motor condition monitoring collects vibration, temperature, and current data using sensors and transmits it wirelessly for analysis. A companion mobile app provides real-time alerts and performance insights. Similar work in [15] and [16] utilizes Raspberry Pi with multiple sensors to create Android-based remote monitoring platforms for induction motors. The significance of doubly fed induction generators for stable operation and maximum power extraction from variable wind sources is highlighted in [17]. Fuzzy logic controllers are shown to reduce torque ripple and improve dynamic performance in precision drive applications, especially in residential contexts, in [18]. Hybrid renewable energy systems tailored for Indian homes, with a focus on sustainability and efficiency, are modeled in [19]. An IoT-driven solution for mobile hybrid renewable systems, emphasizing adaptive control, real-time tracking, and load balancing is discussed in [20].

III. PROPOSED SYSTEM

The proposed system comprises four primary components: WES, BLDC motor, six-step inverter, and an IoT-based monitoring system. The six-step inverter is responsible for converting the DC power generated by the WES into a three-phase

Alternating Current (AC) supply suitable for driving the BLDC motor. The BLDC motor drive system regulates the motor's speed and torque to ensure efficient energy utilization and stable operation. Meanwhile, the IoT monitoring system continuously collects real-time performance data from the motor and associated components, transmitting it to a cloud-based platform for analysis, visualization, and remote control. This architecture allows for effective performance tracking, preventive maintenance, and system optimization. The overall system design is illustrated in Figure 1.

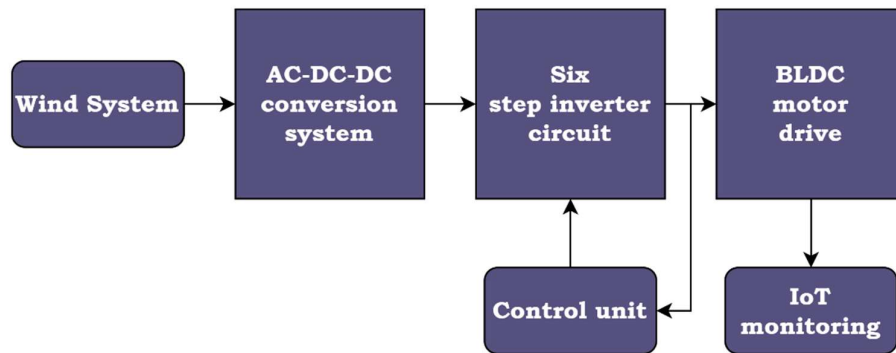


Fig. 1 IoT monitoring of BLDC motor proposed system

The proposed system incorporates both hardware and software components, including a cloud-based IoT platform, sensors, an IoT device, a microcontroller, a control module, a PI speed controller, and a Six-Step Inverter. The PI speed controller plays a critical role in regulating the motor drive by processing real-time feedback on motor speed and generating the appropriate control signals for the Six-Step Inverter. This inverter converts the DC output from the converter into the three-phase AC required to drive the BLDC motor. The IoT-based monitoring system collects operational data from various sensors and transmits it to a cloud platform via a microcontroller-enabled IoT device. The cloud platform features a GUI that enables real-time data visualization, remote monitoring, and control of the motor drive. Key functionalities provided through the GUI include data logging, system performance analytics, and automated warning alerts. Additionally, the platform's control module allows users to send commands back to the controller, ensuring stable and responsive regulation of the motor's speed under varying load and environmental conditions.

A. Speed Control of BLDC motor

The fundamental distinction between BLDC motors and conventional brushed DC motors lies in their construction and control methodology. Unlike brushed DC motors, which rely on mechanical brushes and a commutator for current switching, BLDC motors employ electronic commutation. BLDC motors utilize Hall Effect sensors embedded in the stator to detect the rotor's position, enabling precise control of the motor's rotation. In contrast, traditional DC motors do not incorporate such sensors and depend solely on the mechanical brush-commutator assembly. The Hall sensors in a BLDC motor are aligned with the rotor and play a critical role in determining the timing of commutation signals for effective motor operation. The overall BLDC motor control mechanism [12] is illustrated in Figure 2.

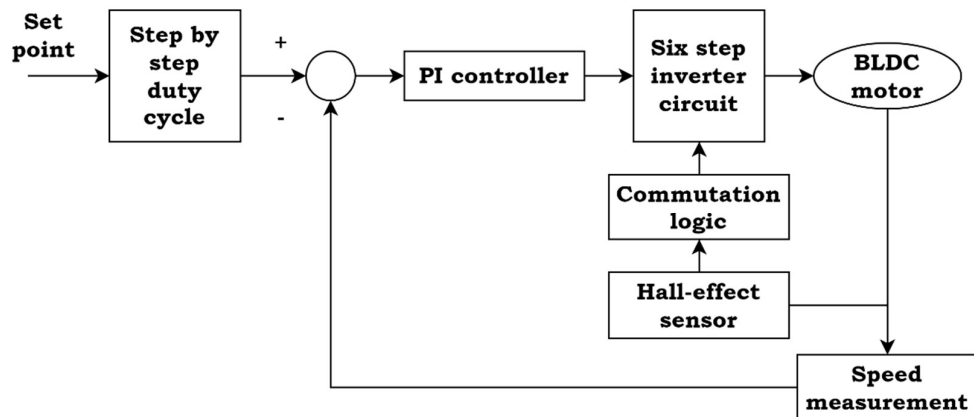


Fig. 2 BLDC motor speed regulation block diagram

BLDC motors can operate under various phase configurations; among them, the three-phase configuration is the most used due to its improved efficiency and better control characteristics. Although three-phase BLDC motors offer superior control precision, they typically produce lower torque and come at a higher cost compared to simpler configurations. In the proposed system, the motor speed is regulated by varying the DC voltage input using a PWM technique. The PWM method adjusts the average voltage supplied to the motor by modulating the duty cycle of the pulses generated by the inverter.

To ensure accurate speed control, a PI speed regulator is implemented. This controller receives feedback on the actual speed of the motor and compares it with reference speed. Based on the error, it generates a control signal that is fed to the six-step inverter, which in turn adjusts the inverter output accordingly. The PI regulator ensures smooth and stable motor operation across different load conditions. Overall, the motor's speed is effectively managed by adjusting either the input voltage or the PWM duty cycle, allowing for efficient and responsive speed regulation.

B. Six Step Inverter Circuit

An inverter is a key power electronic converter that transforms DC into AC, enabling voltage and frequency modulation as required by the load in this case, a BLDC motor. The inverter modulates its AC output by adjusting the DC voltage input or by employing PWM techniques. The term inverter gain refers to the ratio between the inverter's AC output voltage and its DC input voltage. In this work, a six-step inverter configuration is adopted, where six discrete switching devices typically MOSFETs or IGBTs are used to generate three-phase AC output.

The six-step inverter operates by rapidly switching the DC voltage on and off using PWM, with the pulse width being varied to control the average output voltage. The inverter's bridge topology is composed of three legs, each controlling one of the three output phases. Switching the devices in a predefined sequence produces three-phase voltage waveforms that are each 120 degrees out of phase, forming the AC signal required to drive the BLDC motor. Adjusting the switching sequence and duty cycle allows precise control over the motor speed and torque.

A microcontroller interfaces with multiple sensors to acquire real-time feedback on motor parameters such as speed and temperature. Based on this data, the microcontroller issues appropriate switching signals to the inverter. The system is also integrated with an IoT module such as the ESP32 which enables cloud

connectivity. This IoT integration allows for remote monitoring, control, and data analysis via a cloud-based platform. Control commands issued through the platform are relayed to the microcontroller, which then modulates the inverter operation accordingly to regulate the motor's speed and ensure optimal performance.

IV. RESULTS AND DISCUSSIONS

The proposed system was developed, implemented, and systematically evaluated to assess its operational efficiency. The findings confirm that the IoT-based monitoring framework successfully enabled real-time observation and control of the wind-powered BLDC motor drive. Critical parameters such as motor speed, voltage, and wind speed were accurately captured by the system's integrated sensors. The microcontroller processed this data and, through a wireless communication module, transmitted it to the IoT device. The IoT device subsequently forwarded the information to a cloud-based platform for storage, visualization, and further analysis.

The modeling and experimental evaluation of the PI speed controller demonstrated its robust regulation capability and stable performance. The PI regulator effectively controlled the speed of the BLDC motor, ensuring rapid convergence to the desired reference speed. Specifically, the motor achieved the target speed of 1800 rpm within 0.05 seconds, indicating a fast dynamic response. This performance was continuously monitored and validated in real-time via the IoT-based system. The speed tracking response is illustrated in Figure 3. The simulation block diagram of the proposed system is depicted in Figure 4.

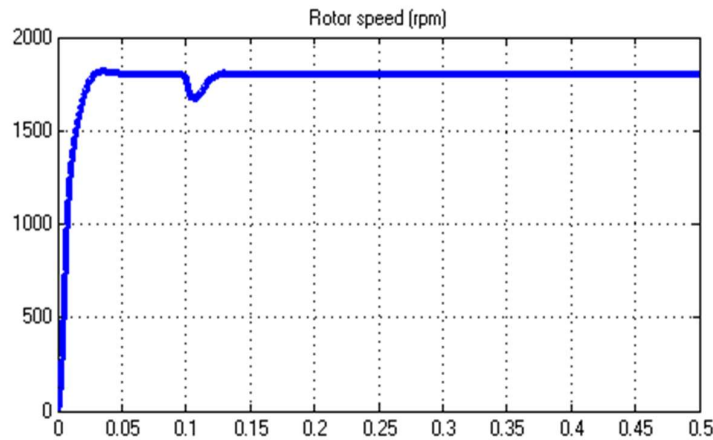


Fig. 3 IoT monitored motor speed

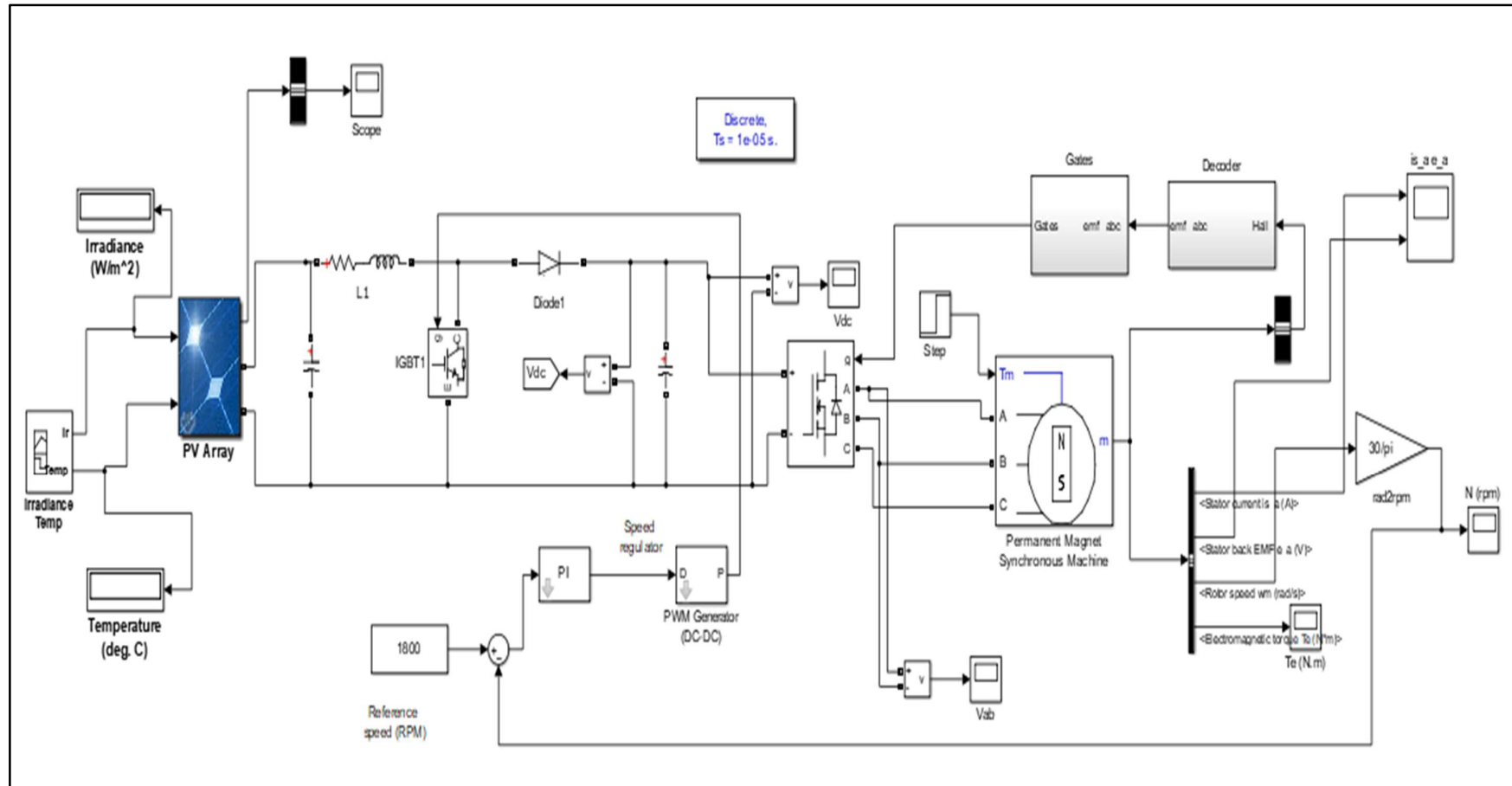


Fig. 4 Simulation model of wind-based BLDC motor drive

The electromagnetic torque generated by the BLDC motor drive initially peaks at approximately 16 Nm and stabilizes to a steady-state value of 4 Nm within 0.1 seconds, demonstrating a rapid and well-damped transient response. This torque behavior is illustrated in Figure 5.

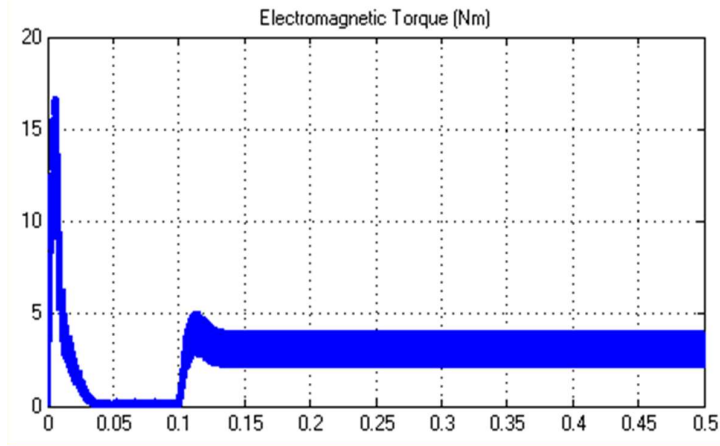


Fig. 5 BLDC motor torque

The six-step inverter regulated the BLDC motor's speed based on the output signal from the PI controller. The motor's speed response was continuously monitored and compared against the reference speed. The results demonstrated that the motor accurately tracked the reference speed with minimal steady-state error, validating the effectiveness of the PI controller's tracking capability. The output voltage from the six-step inverter reached 300 V and stabilized within 0.1 seconds, as illustrated in Figure 6.

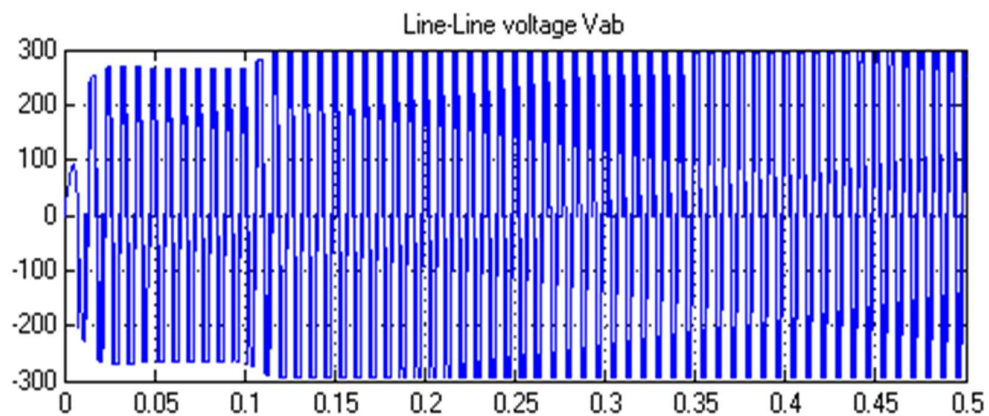


Fig. 6 Six Step inverter Voltage

Continuous monitoring of the wind-driven BLDC motor drive provided several key advantages. The system effectively detected anomalies and generated timely alerts, enabling rapid maintenance and fault rectification. Additionally, it

dynamically adjusted the motor speed based on real-time wind speed variations to optimize energy efficiency. The integrated IoT functionality also allowed for remote monitoring and control, making the solution particularly well-suited for deployment in remote or unmanned locations.

V. CONCLUSIONS

The proposed IoT-based wind-powered BLDC motor drive system, incorporating a six-step inverter and PI speed controller, demonstrates an efficient solution for real-time speed tracking, remote monitoring, and enhanced system reliability. The IoT monitoring system enabled live data acquisition, visualization, and remote control, while the PI controller ensured rapid and accurate speed tracking with minimal deviation from the reference. The integration with a cloud-based IoT platform further facilitated remote system management, making the solution highly suitable for deployment in isolated or unmanned environments. Future research could focus on improving system efficiency through the integration of advanced control algorithms such as model predictive control, adaptive fuzzy logic, or AI-based optimization techniques. Incorporating fault detection and diagnostic capabilities would also enhance system robustness by enabling automated identification and mitigation of potential failures. Moreover, the proposed system architecture can be scaled and adapted for use with other RES such as solar or hybrid configurations, to further improve sustainability and energy utilization.

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Conflicts of Interest: The authors declare that they have no conflicts of interest to report regarding the present study.

REFERENCES

- [1]. T. Das, S. Halder and V.A. Shah, "Efficient energy management system for open-winding motor drive for electrical vehicle with hybrid energy storage systems," *IEEE Global Conference on Computing, Power and Communication Technologies*, 2022, pp. 1-7.
- [2]. R.R. Nair and G. Narayanan, "Emulation of wind turbine system using vector controlled induction motor drive," *IEEE International Conference on Power Electronics, Drives and Energy Systems*, vol. 56, no. 4, 2020, pp. 4124-4133.
- [3]. C. Xia, Z. Li, and T. Shi, "A control strategy for four-switch three-phase brushless DC motor using single current sensor," *IEEE Transactions on Industrial Electronics*, vol. 56, no. 6, 2009, pp. 2058-2066.
- [4]. P.K. Sharma and A.S. Sindekar, "Performance analysis and comparison of BLDC motor drive using PI and FOC," *International Conference on Global Trends in Signal Processing, Information Computing and Communication*, 2016, pp. 485-492.
- [5]. M. Masmoudi, B.E. Badsı and A. Masmoudi, "DTC of B4-inverter-fed BLDC motor drives with reduced torque ripple during sector-to-sector commutations," *IEEE Transactions on Power Electronics*, vol. 29, no. 9, 2014, pp. 4855-4865.

- [6]. Y. Lee and J. Kim, "Analysis of the three-phase inverter power efficiency of a BLDC motor drive using conventional six-step and inverted pulsewidth modulation driving schemes," *Canadian Journal of Electrical and Computer Engineering*, vol. 42, no. 1, 2019, pp. 34-40.
- [7]. K. Oguchi, A. Kawaguchi, T. Kubota and N. Hoshi, "A novel six-phase inverter system with 60-step output voltages for high-power motor drives," *IEEE Transactions on Industry Applications*, vol. 35, no. 5, 1999, pp. 1141-1149.
- [8]. M.L. Hossain, A. Abu-Siada, S.M. Mueen, M.M. Hasan and M.M. Rahman, "Industrial IoT-based condition monitoring for wind energy conversion system," *Journal of Power and Energy Systems*, vol. 7, no. 3, 2021, pp. 654-664.
- [9]. S. Ghosh, "Neuro-fuzzy-based IoT assisted power monitoring system for smart grid," *IEEE Access*, vol. 9, 2021, pp. 168587-168599.
- [10]. R. Saranya, R. Punithavalli, E. Nandakumar and R. Priya, "Web monitoring and speed control of solar based bldc motor with IoT," *International Conference on Advanced Computing and Communication Systems*, 2019, pp. 808-812.
- [11]. M.R. Haque and S. Khan, "The modified proportional integral controller for the BLDC motor and electric vehicle," *IEEE International IoT, Electronics and Mechatronics Conference*, 2021, pp. 1-5.
- [12]. R. Rakhmawati, F.D. Murdianto and M. Wildan Alim, "Soft starting and performance evaluation of PI speed controller for brushless DC motor using three phase six step inverter," *International Seminar on Application for Technology of Information and Communication*, 2018, pp. 121-126.
- [13]. T.A. Khan, F.A. Shaikh, S. Khan and M.F. Siddiqui, "Real-time wireless monitoring for three phase motors in industry: A cost-effective solution using IoT," *IEEE International Conference on Smart Instrumentation, Measurement and Application*, 2019, pp. 1-5.
- [14]. D. Shyamala, D. Swathi, J.L. Prasanna and A. Ajitha, "IoT platform for condition monitoring of industrial motors," *International Conference on Communication and Electronics Systems*, 2017, pp. 260-265.
- [15]. N. Dehbashi, M. SeyyedHosseini and A. Yazdian-Varjani, "IoT based condition monitoring and control of induction motor using raspberry Pi," *Power Electronics, Drive Systems, and Technologies Conference*, 2022, pp. 134-138.
- [16]. M. Errouha, Q. Combe, N.E. Ouanjli and S. Motahhir, "A review of modern techniques for efficient control of AC motors utilized in PV water pumping system," *Irrigation Science*, 2024, pp. 1-21.
- [17]. M. Salman, S.A. Kashif, M.S. Fakhar, A. Rasool and A.S. Hussien, "Optimizing power generation in a hybrid solar wind energy system using a DFIG-based control approach," *Scientific Reports*, vol. 15, no. 1, 2025, pp. 1-30.
- [18]. A. Tikkani, R.R. Karasani, K.V.G. Rao, C.N. Kalyan, B.S. Goud, M. Bajaj and M.B. Tuka, "Fuzzy-2 deployment in indirect vector control and hybrid space vector modulation for a two-level inverter fed induction motor drive," *Scientific Reports*, vol. 15, no. 1, 2025, pp. 1-25.
- [19]. A. Mohanasundaram and P. Valsalal, "Design of a wind-solar hybrid energy air conditioning system using BLDC motor for the Indian home environment," *Electrical Engineering*, vol. 105, no. 3, 2023, pp. 1717-1728.
- [20]. A. Yadav, N. Priyadarshi, F. Azam, R. Singh and A. Gehlot, "An IoT based intelligent energy management of PV/wind hybrid renewable energy system for transportation applications," *AIP Conference Proceedings*, vol. 2771, 2023, no. 1, 2023, pp. 1-9.